USN. | | | | | | | | | | [22MTASPDE322]

THIRD SEMESTER M. TECH EXAMINATION, DECEMBER-2024

RAMJET AND SCRAMJET PROPULSION

Time: 3 Hours **Maximum Marks: 100**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
- **ii.** Missing data may be suitably assumed.
- **iii.** Use of gas table is permitted

1.	**a.**	A ramjet is flying at Mach 2.0 at an altitude where $T_0 = -50°C$ and the engine airflow rate is 10 kg/s. If the exhaust Mach number of the ramjet is equal to the flight Mach number, i.e., $M_9=M_0$, with perfectly expanded nozzle and $T_9 = 2500$ K, calculate (a) the engine ram drag D_{ram} in kN (b) the nozzle gross thrust F_B in kN (c) the engine net thrust F_n in kN (d) and the engine propulsive efficiency η_p. Assume gas properties remain the same throughout the engine, i.e., $\gamma=1.4$ and $C_p=1004$ J/kg·K. Also, assume that the fuel flow rate is 4% of airflow rate.	**CO1**	**L4**	**10**
	b.	A Ramjet engine operates on a simple ideal Brayton cycle with a pressure ratio of 10. Heat is added to the cycle at a rate of 500 kW; air passes through the engine at a rate of 1 kg/s; and the air at the beginning of the compression is at 70 kPa and 0°C. Determine the power produced by this engine and its thermal efficiency. Use constant specific heats of at room temperature.	**CO1**	**L3**	**10**
2.	**a.**	Illustrate the three types of supersonic inlets with its merits and demerits.	**CO2**	**L2**	**10**
	b.	Explain the factors that contribute to the unstart of a supersonic inlet.	**CO2**	**L2**	**10**
3.	**a.**	Illustrate the combustion process in a RAMJET engine.	**CO3**	**L2**	**10**
	b.	Explain the classification of SCRAMJET engines.	**CO4**	**L2**	**10**
		OR			
4.	**a.**	Describe the methods used to inject the fuel in a typical supersonic combustion process.	**CO4**	**L2**	**10**
	b.	Explain the main challenges in achieving stable combustion at high speeds in a RAMJET engine.	**CO3**	**L2**	**10**

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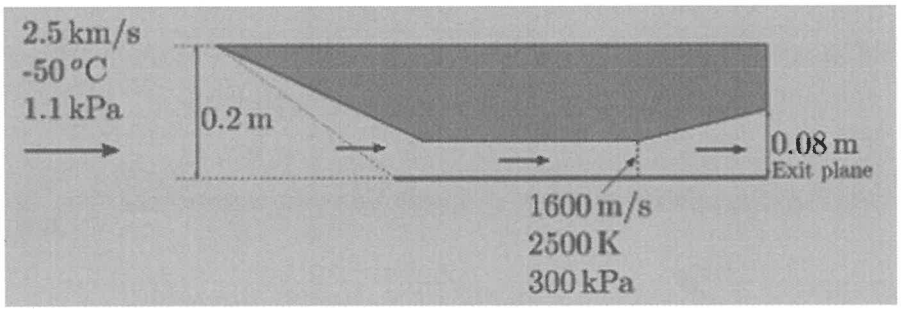
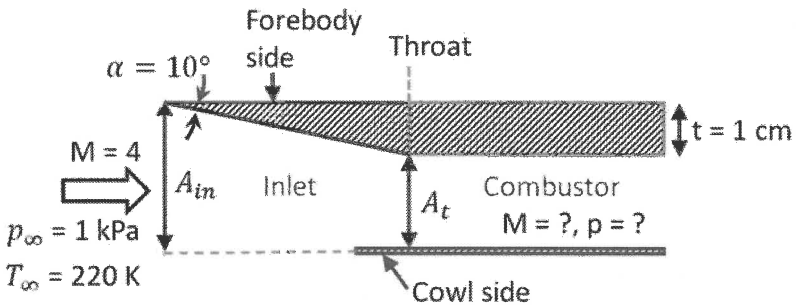
5.	**a.**	Explain the mechanisms that govern the fuel-air mixing process in SCRAMJET engines.	**CO5**	**L2**	**10**
	b.	Describe the air-fuel mixing in a supersonic combustor using turbulent shear layer mixing process.	**CO5**	**L2**	**10**
		OR			
6.	**a.**	Elaborate the effect of area variation on heat addition in a supersonic combustor.	**CO5**	**L2**	**10**
	b.	Explain the need and function of isolator in SCRAMJET engine.	**CO5**	**L2**	**10**
7.	**a.**	A SCRAMJET is flying at an altitude of 30 km, where the air temperature is 50°C and pressure is 1.1 kPa, and at a speed of 2.5 km/s. In the combustor, hydrogen fuel is added at a rate of 3 kg of hydrogen per 100 kg of air. As a result of the combustion process, the mixture at the exit of the combustor is now at 2500 K, 300 kPa with a velocity of 1600 m/s. From this point, the flow is expanded through a nozzle to the exit plane with a density ratio of 11.35 ($\rho_{combustor}/\rho_{exit}$ = 11.35). The SCRAMJET shown in Figure-1 is a 2D geometry, that is, from front view it looks rectangular with a width of 500 mm. You can assume that the ideal gas relations can be used at each point in the flow and use the specific gas constant for air. Determine: (i) Mass flow entering the engine (ii) Nozzle exit velocity (iii) Thrust produced.  **Figure-1**	**CO6**	**L4**	**15**
	b.	Describe the factors limiting the maximum allowable compression temperature in a SCRAMJET engine.	**CO6**	**L2**	**05**
		OR			
8.	**a.**	Consider a simple model scramjet engine shown in Figure-2 consisting of an inlet, followed by a combustor. At cruise, the inlet is designed to provide a wave free (no shock, no expansion fan) flow into the combustor. The fore body of the inlet can be treated as a wedge with a turning angle of α=10∘. This inlet is designed such that it compresses the incoming flow using exactly two finite	**CO6**	**L4**	**15**

		strength oblique shock waves (i.e., Mach waves are not considered as oblique shock waves). Note that the cowl side is treated as an infinitely thin plate placed at 0∘ to the freestream flow. Assume a 2D inlet with a unit span in the out of plane direction. The combustor is modeled as a constant area duct. (i) Sketch the placement of the cowl side surface and the oblique shocks to deliver a wave free flow into the combustor. (ii) What is the physical contraction ratio of the inlet (A_{in} /A_t)? (iii) What are the Mach number and pressure delivered to the combustor? (iv) Based on the answer for (iii) argue if this engine can operate efficiently in SCRAMJET mode or if it needs to be operated in a ramjet mode?  **Figure-2**			
	b.	Explain the Influence of cycle static temperature ratio on the SCRAMJET engine performance.	**CO6**	**L2**	**05**